TECH NEXUS

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PROBLEM

WHY IS CHOOSING A TECH GADGET DIFFICULT?

- The tech market offers a large number of gadgets with similar features and prices.
- Users often struggle to compare products and identify what best suits their needs.
- Most existing tools rely mainly on price-based filtering, ignoring usage and priority features.
- This leads to confusion and decision delay, commonly known as analysis paralysis.
- Hence, a smart multi-parameter recommendation system is required.

OUR SOLUTION

- Tech-Nexus is a smart, logic-based recommendation system for tech gadgets.
- It helps users select the most suitable product instead of showing too many options.
- The system considers multiple parameters such as budget, usage type, and priority features.
- It works like a virtual decision assistant, guiding users step-by-step.
- The final output is a well-matched recommendation tailored to the user's needs.

SYSTEM CONCEPT

- The system follows a decision-tree and rule-based logic model.
- Each input acts as a decision node.
- Products failing to satisfy conditions are removed dynamically.
- Logical sequencing ensures progressive filtering.
- This structured elimination process increases system reliability.
- The final output is optimized for relevance and accuracy.

USER INPUT PARAMETERS

The system requires multiple inputs from the user to understand their exact needs:

- Product Category: Defines the domain such as smartphones and laptops
- Budget Constraint: Filters products within the user's financial limit.
- Usage Type: Determines functional requirements such as gaming, photography, office work, or general use.
- Priority Feature: Further refines selection based on user preference, such as battery life, performance, or portability.

Together, these parameters form a complete user profile, allowing personalized and meaningful recommendations.

SYSTEM WORKFLOW

1. Initialization

Start: Program launches and loads the database.
Main Menu: User selects a category:
1. Find a Laptop 2. Find a Phone 3. Exit

2. User Input Phase (Data Collection)

If Laptop Selected:

- Budget
- Usage Preference: Gaming (High Performance), Student (Battery/Portability), Professional (Coding/Work)
- Brand preference

If Phone Selected:

- Budget
- Priority Feature: Camera Quality, Battery Life, Performance
- OS Preference: Android vs. iOS.
- Brand preference

3. Processing Phase (The Algorithm)

System iterates through the database array.
Filters: Checks every device against User Inputs (Price <= Budget AND Usage == Preference).
Exclusion: Skips empty slots and non-matching devices.

4. Output & Termination

Success: Displays a formatted list of matching devices with specs (RAM, Storage, Price).
Failure: If count == 0, displays a "No devices found" message.
End: Program terminates or returns to Main Menu.

```c
switch(ch) {
    case 1:
        recommendLaptop(laptopDB);
        break;

    case 2:
        recommendPhone(phoneDB);
        break;

    case 3:
        printf("\nExiting");
        break;

    default:
        printf("\nInvalid choice! Try again");
}
```

```c
p[10].id = 11;
strcpy(p[10].name, "Apple iPhone 17 Pro");
p[10].price = 134900;
p[10].brand = 7;      // 7=Apple
p[10].os = 2;        // 2=iOS
p[10].ram = 12;
p[10].camera = 48;
p[10].storage = 256;
p[10].battery = 3988;
p[10].usage = 1;
```

```
l[11].id = 12;
strcpy(l[11].name, "ASUS TUF F16");
l[11].price = 97000;
l[11].brand = 1;    // ASUS = 1
l[11].usage = 1;   // 1 = gaming
l[11].ram = 16;
l[11].storage = 512;
```

```
if (db[i].price <= l_budget && (db[i].usage == l_usage || l_usage == 0) && (db[i].brand == brand_prefference || brand_prefference == 0)) {
    printLaptopRow(db[i]);
    count++;
```

SMART FILTERING & DECISION LOGIC

The filtering mechanism is implemented using compound logical expressions. The system primarily uses AND (&&) operators to ensure that all mandatory conditions are satisfied. OR (||) operators may be used to allow flexibility when multiple features are acceptable.

Example logic:

- Product price ⩽ user budget
- AND usage type matches user requirement
- AND priority feature meets minimum threshold

Any product that fails to satisfy even one condition is removed. This logical approach ensures precision and relevance in recommendations.

TECHNICAL IMPLEMENTATION

The system is implemented using the C programming language. Product information is stored using structures (struct), allowing multiple attributes to be grouped under a single product entity.

Arrays of structures are used to maintain the product database. Loops traverse the dataset efficiently, while conditional statements apply the filtering logic. Logical operators ensure correct decision-making.

The program demonstrates structured programming, data abstraction, and effective use of control flow mechanisms.

ADVANTAGES AND INNOVATION

Tech-Nexus offers several advantages over traditional tech recommendation tools by using a multi-parameter and logic-driven approach. Unlike basic systems that rely only on price comparison, this system evaluates user requirements holistically, leading to more accurate and meaningful recommendations.

One of the major advantages is the reduction of decision fatigue and analysis paralysis. By eliminating unsuitable products at each stage, the system minimizes the cognitive load on users and saves significant time during the selection process.

From a technical perspective, Tech-Nexus demonstrates advanced logical reasoning through the use of compound conditions, decision-tree modeling, and structured filtering. The implementation showcases practical application of programming concepts such as data structures, conditional logic, and iterative processing.

The system is also scalable and adaptable. New products, additional features, or more categories can be integrated without altering the core logic. This makes Tech-Nexus suitable for real-world deployment and future expansion.
Overall, Tech-Nexus bridges the gap between user needs and complex technical specifications, making it an innovative, practical, and intelligent recommendation system.

CONCLUSION

Tech-Nexus successfully demonstrates how structured programming logic can be applied to solve a real-world problem. By using a multi-parameter, decision-based approach, the system simplifies the complex process of selecting technology products and transforms it into a clear and systematic workflow.
The project highlights the effective use of core programming concepts such as structures, loops, conditional statements, and logical operators. Instead of relying on random comparisons or single-parameter filtering, Tech-Nexus ensures that recommendations are accurate, relevant, and aligned with user requirements.
By reducing confusion and decision fatigue, the system improves user confidence and enhances the overall decision-making experience. The logical elimination process ensures that only the most suitable products are recommended, making the system both practical and efficient.

Future Scope:

- Integration of user ratings and reviews to enhance recommendation accuracy
- Implementation of machine learning algorithms for personalized suggestions
- Deployment as a web or mobile-based application
- Expansion to additional product categories and real-time data integration

In conclusion, Tech-Nexus serves as a scalable, intelligent, and practical recommendation system with strong potential for future enhancements and real-world application.

THANK YOU